Course Introduction, Framework for Analyzing IT Trends, Global IT Landscape

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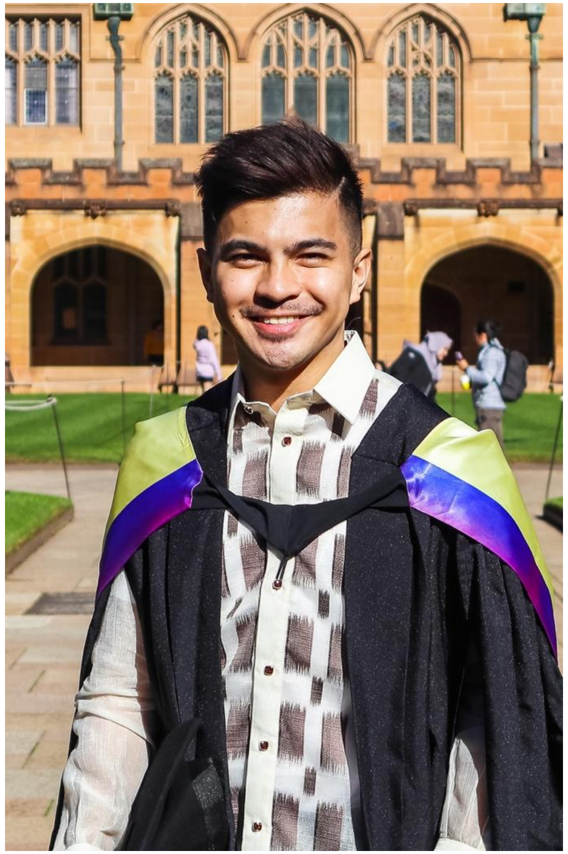
About the Lecturer

- About 11 years in the IT industry
- Current works as Senior AI Project Manager at Aboitiz Data Innovation, Lecturer at PUP & MSEUF, Start-up Founder
- Areas of Interest: Project Management and Business Development
- Master of IT Management at The University of Sydney, Australia
- BS Computer Science at MSEUF-Lucena



Class Schedule

- Alternate weekly online sessions in 18 weeks (about 4 months)
- Flexible class schedule based on specified schedule every Saturday (wait for further notice a day before class)
- Online meeting links will be sent to group chat prior classes

Class Reminders

- If not possible to attend the class/es, inform lecturer in the group chat
- Email questions via: dkqbulawan@pup.edu.ph (avoid private messages) or use the group chat

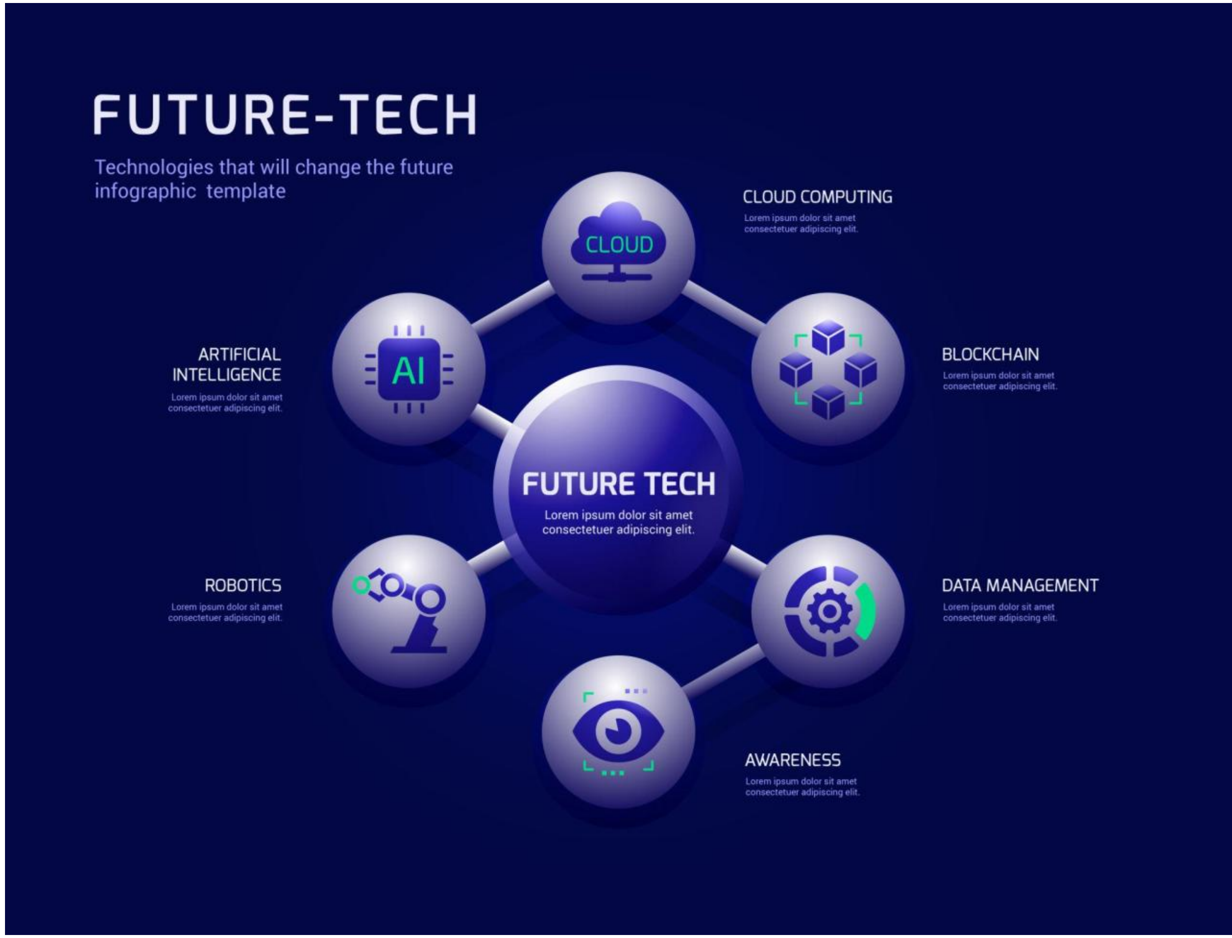
Course Map

Week Number	Inclusive Dates	Module Number	Assessments	Topics
1		1	-	Course Introduction, Discussion of Deliverables, Framework for Analyzing IT Trends, Global IT Landscape
2-3		2	Analysis Paper (I)	Artificial Intelligence (AI) & Generative AI, Large Language Models, Enterprise AI Adoption, AI Ethics, Bias and Explainability, AI Regulation (EU AI Act, PH AI initiatives)
4-5		3	Presentation (G)	Cybersecurity Trends, Zero Trust Architecture, Ransomware Evolution, Cloud-Native Architecture, Multi-Cloud Strategy, Edge Computing
6-7		4	Review Paper (I)	Data Governance, Data Privacy (Philippine DPA updates), Data Ethics & Responsible Analytics, Blockchain & Web3, Smart Contracts, Enterprise Adoption vs Speculation
8		5	Plan Report (G)	Internet of Things (IoT), Smart Systems & Industry 4.0, IoT Security Challenges
9		-	Exam	**MIDTERM EXAMINATION**
10-11		6	Research Paper (I)	Automation & Hyperautomation, RPA & Process Mining, Low-Code / No-Code Platforms, Quantum Computing, Post-Quantum Cryptography
12-13		7	Proposal Paper (G)	Sustainable IT, Green Computing, ESG & Digital Responsibility, DevOps & DevSecOps, Platform Engineering, GitOps
14-15		8	Analysis Report (I)	Extended Reality (AR/VR/MR), Spatial Computing, Human-Computer Interaction Trends, Digital Transformation in the Philippines, eGovernment, FinTech & Digital Banking
16-17		9	Presentation (G)	IT Project Failures, Governance Breakdowns, Ethical and Strategic Risks
18		-	Exam	**FINALEXAMINATION**

Learning Objectives

In the end of this session, students will be able to:

- Explain the purpose and structure of ITEC 304
- Define what constitutes an “IT trend”
- Differentiate hype from sustainable innovation
- Identify major global technology drivers
- Recognize the role of IT professionals in shaping digital transformation



What is a Technology Trend?

Technology trends are patterns of technological change that show sustained growth and impact over time. A technology trend:

- Shows consistent industry adoption
- Impacts multiple sectors
- Evolves beyond experimental phase
- Influences skills demand

This course examines **contemporary issues, emerging technologies, and disruptive trends** shaping the field of Information Technology. It provides students with analytical frameworks to evaluate technological developments, assess their societal and organizational impacts, and articulate informed positions on complex digital challenges.

Technology Issue vs Technology Trend

Not all trends are positive. Some create risks and ethical concerns.
Trend = Growth direction
Issue = Problem or challenge created by tech

Examples:

- AI (Trend) → Bias and misinformation (Issue)
- Cloud computing (Trend) → Data breaches (Issue)

Why This Course Matters?

The IT industry changes faster than textbooks.

Reasons this course is important:

- Technologies evolve every 6–12 months
- Skills become obsolete quickly
- Employers value analytical thinkers
- Policy and regulation affect IT careers

Global IT Landscape

The past 3 years accelerated digital transformation globally.

Major global shifts:

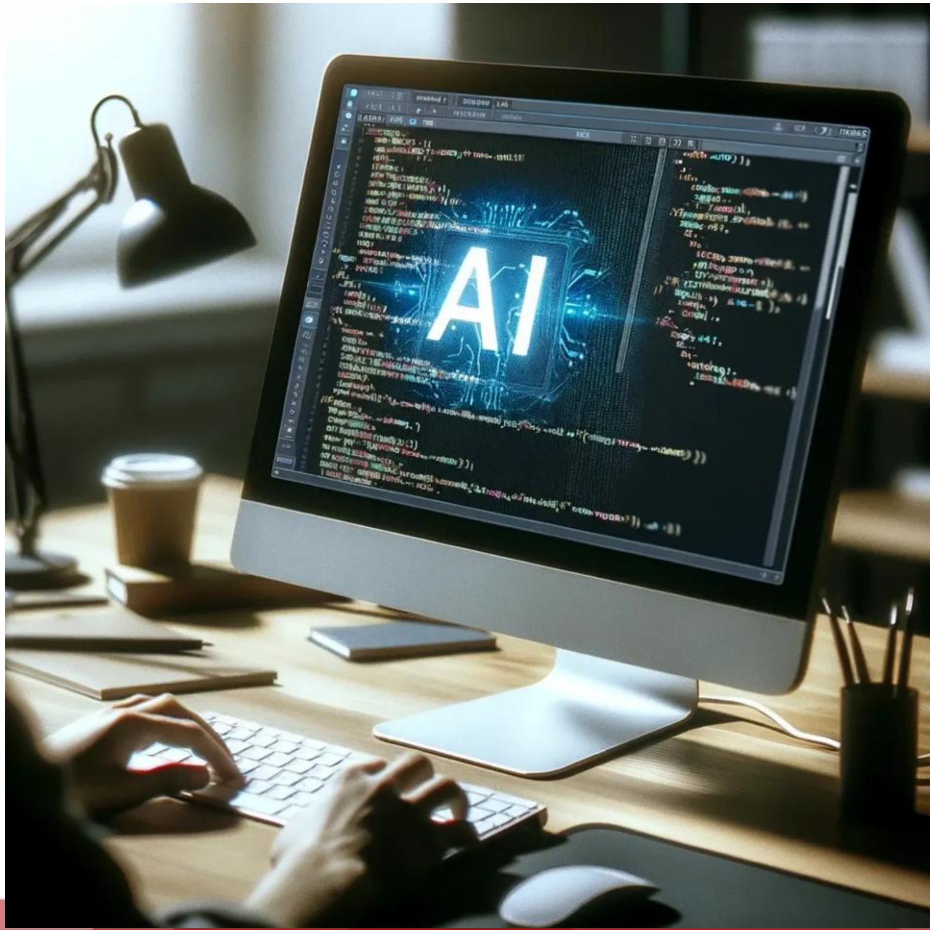
- AI mainstream adoption
- Cloud-first enterprises
- Rise of cybersecurity threats
- Automation of white-collar jobs

Technology Issue vs Technology Trend

Generative AI changed how we work and create content.

Key developments:

- Large Language Models
- AI copilots in software development
- AI-generated media
- Enterprise AI governance

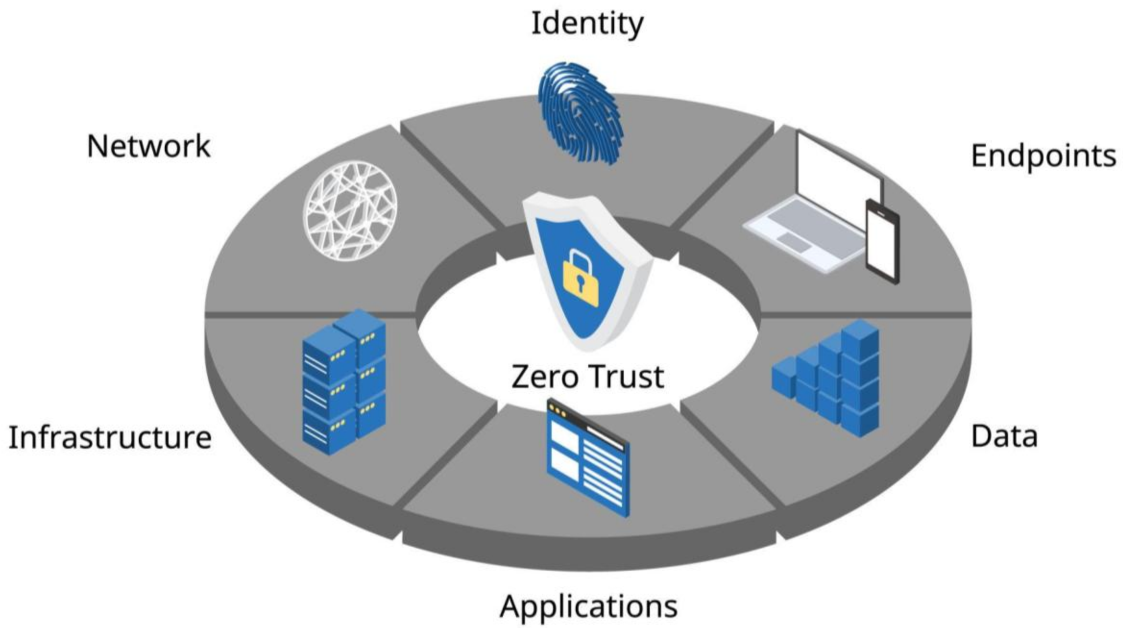


Cybersecurity Arms Race

As technology advances, attacks also become sophisticated.

Current realities:

- Ransomware-as-a-Service
- AI-powered phishing
- Zero Trust Architecture
- Cloud vulnerabilities

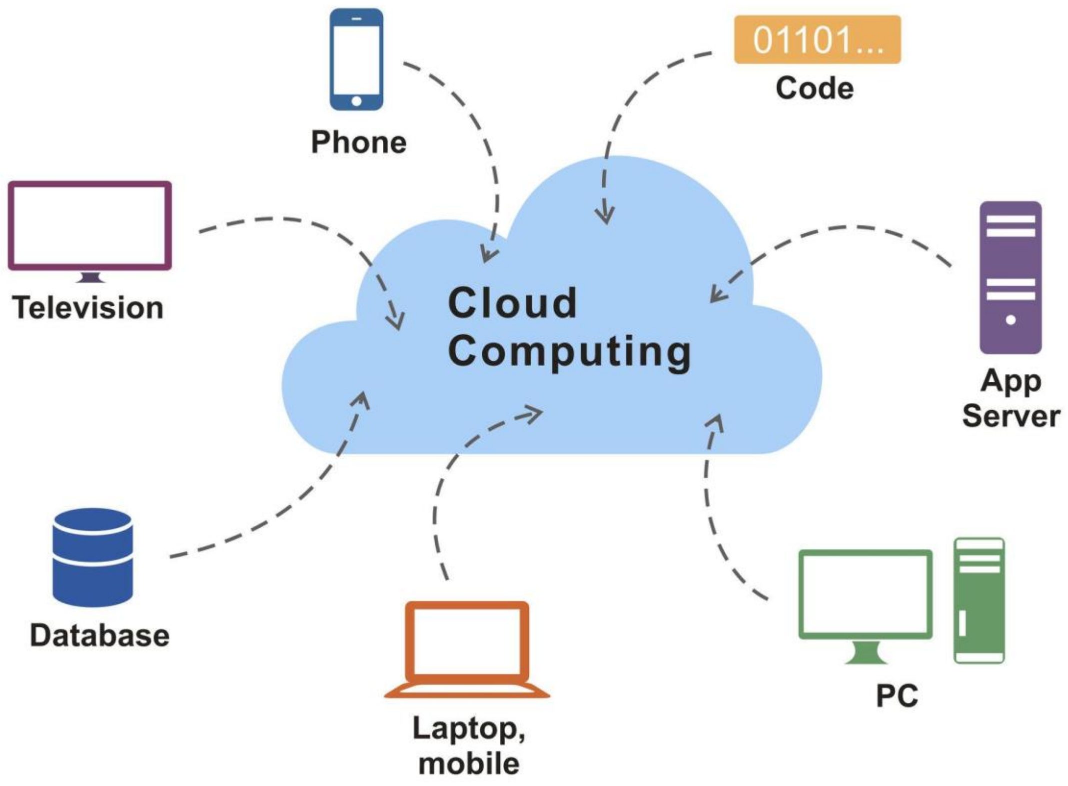


Cloud & Platform Dominance

Cloud computing is now infrastructure backbone.

Trends:

- Multi-cloud strategy
- Serverless computing
- Platform engineering
- DevOps automation

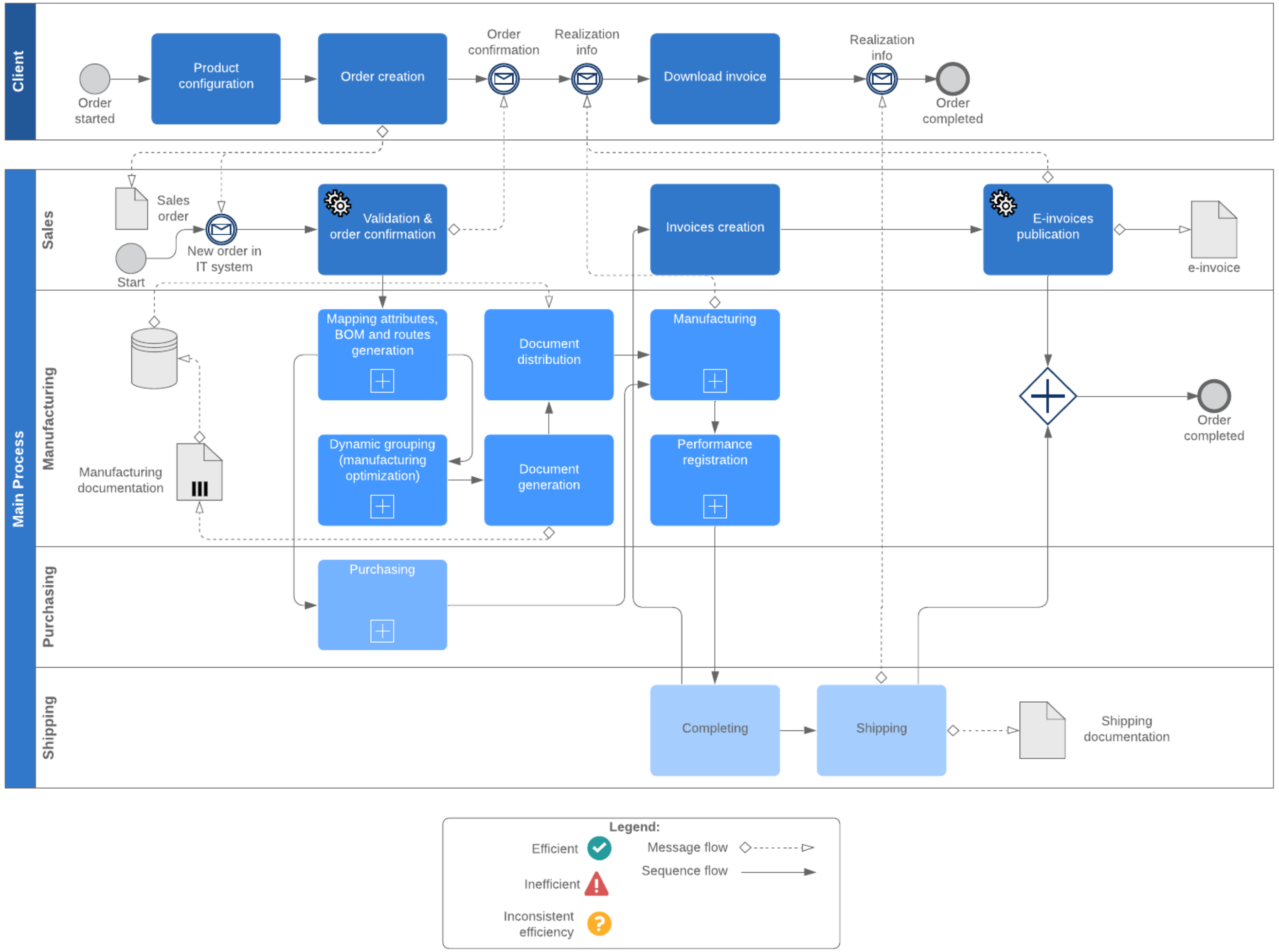


Automation & Jobs

Automation is reshaping labor markets

Impact areas:

- Process automation
- AI-assisted coding
- Customer service automation
- Data analytics automation



Philippine Digital Landscape

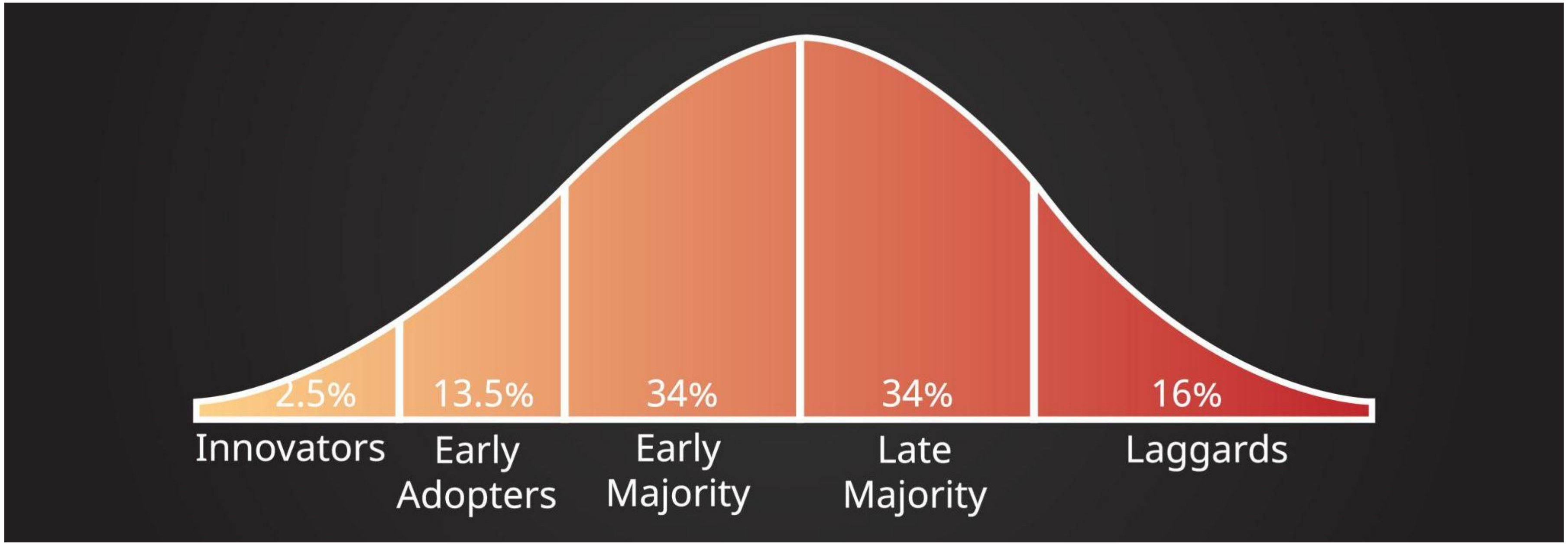
Digital transformation is not just global — it affects us locally.

Philippine context:

- eGovernment initiatives
- Digital banking growth
- Data privacy enforcement
- IT outsourcing competitiveness

The Technology Lifecycle

Not all technologies survive. Some grow. Some disappear quietly.
Understanding the lifecycle helps us avoid chasing hype.



Hype vs Real Transformation

Popularity does not equal long-term impact.
We need to separate marketing noise from structural change.

Hype Indicators:

- Heavy media coverage
- Investor excitement
- Limited practical use

Real Transformation Indicators:

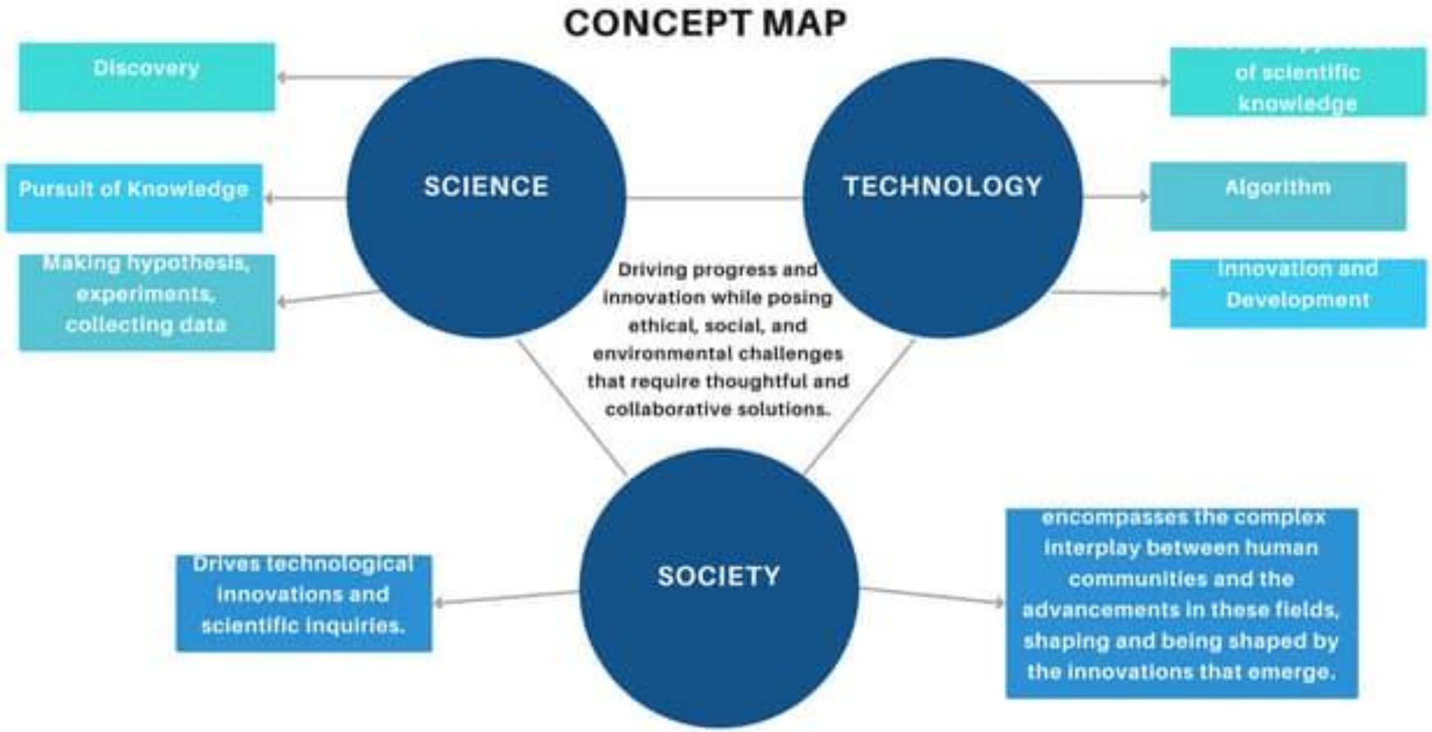
- Industry integration
- Policy/regulation alignment
- Skills demand increase

The Role of Ethics in IT Trends

Technology decisions are never purely technical.
Every system affects real people.

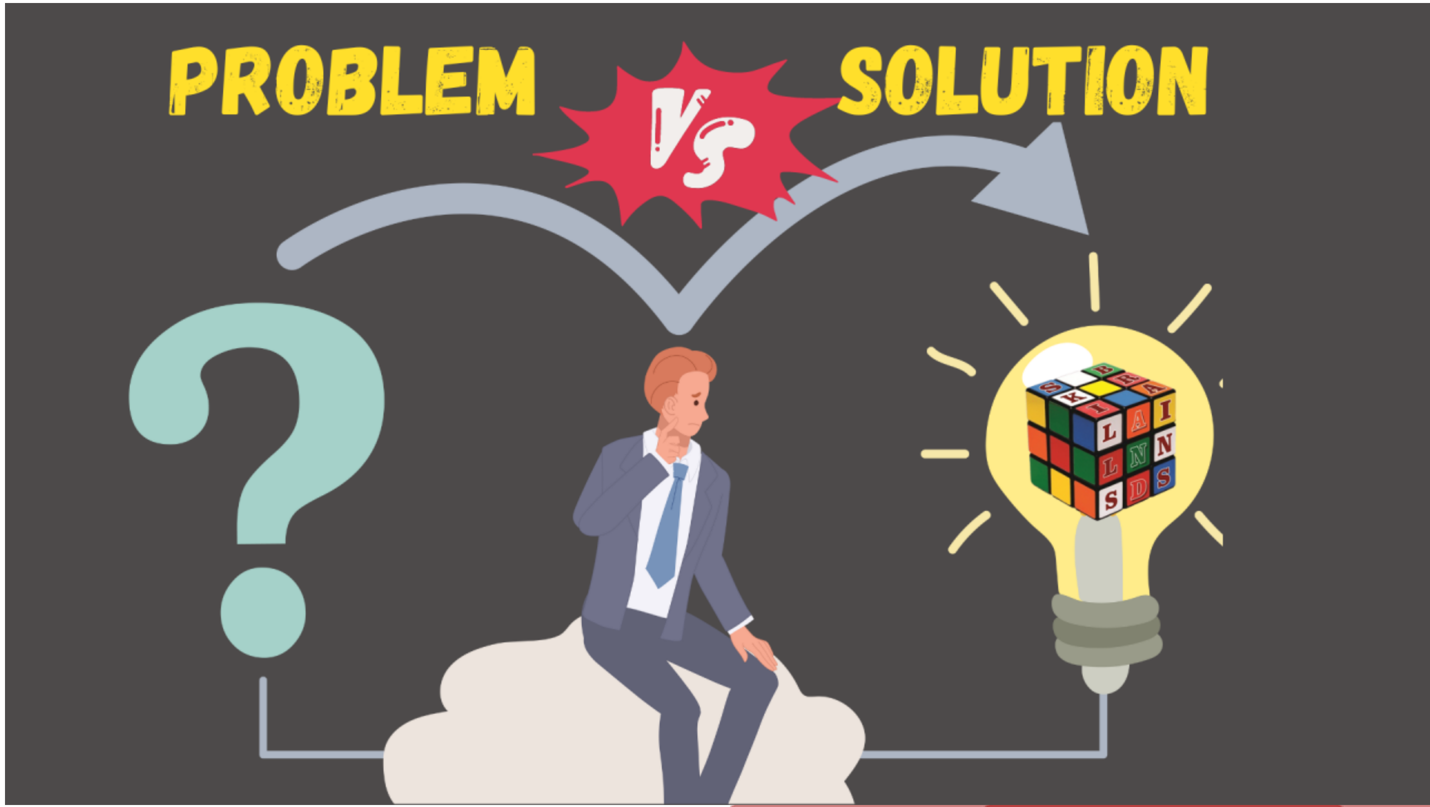
Ethical Dimensions:

- Data privacy
- Algorithmic bias
- Job displacement
- Environmental impact



Critical Thinking in IT

This course is a thinking course. You are not here to memorize — you are here to analyze.



Critical Thinking Skills:

- Ask “Who benefits?”
- Ask “What risks exist?”
- Check data sources
- Compare multiple perspectives

Seminar Expectations

This is not a passive lecture class.

You are expected to engage at a professional level.

Expectations:

- Research-based arguments
- Clear structured presentations
- Respectful debate
- Evidence-backed opinions

Sources of Credible Tech Research

Not all online content is reliable.
You must learn where professionals get information.

Recommended Sources:

- Gartner
- McKinsey Digital
- MIT Technology Review
- IEEE Spectrum
- WEF Reports
- Government policy documents

The IT Professional of 2026

The future IT professional is hybrid.

Technical + analytical + ethical.

Future-Ready Skills:

- Systems thinking
- AI literacy
- Cyber awareness
- Policy awareness
- Communication skills

How You Will Be Evaluated?

Evaluation is based on thinking depth, not memorization.

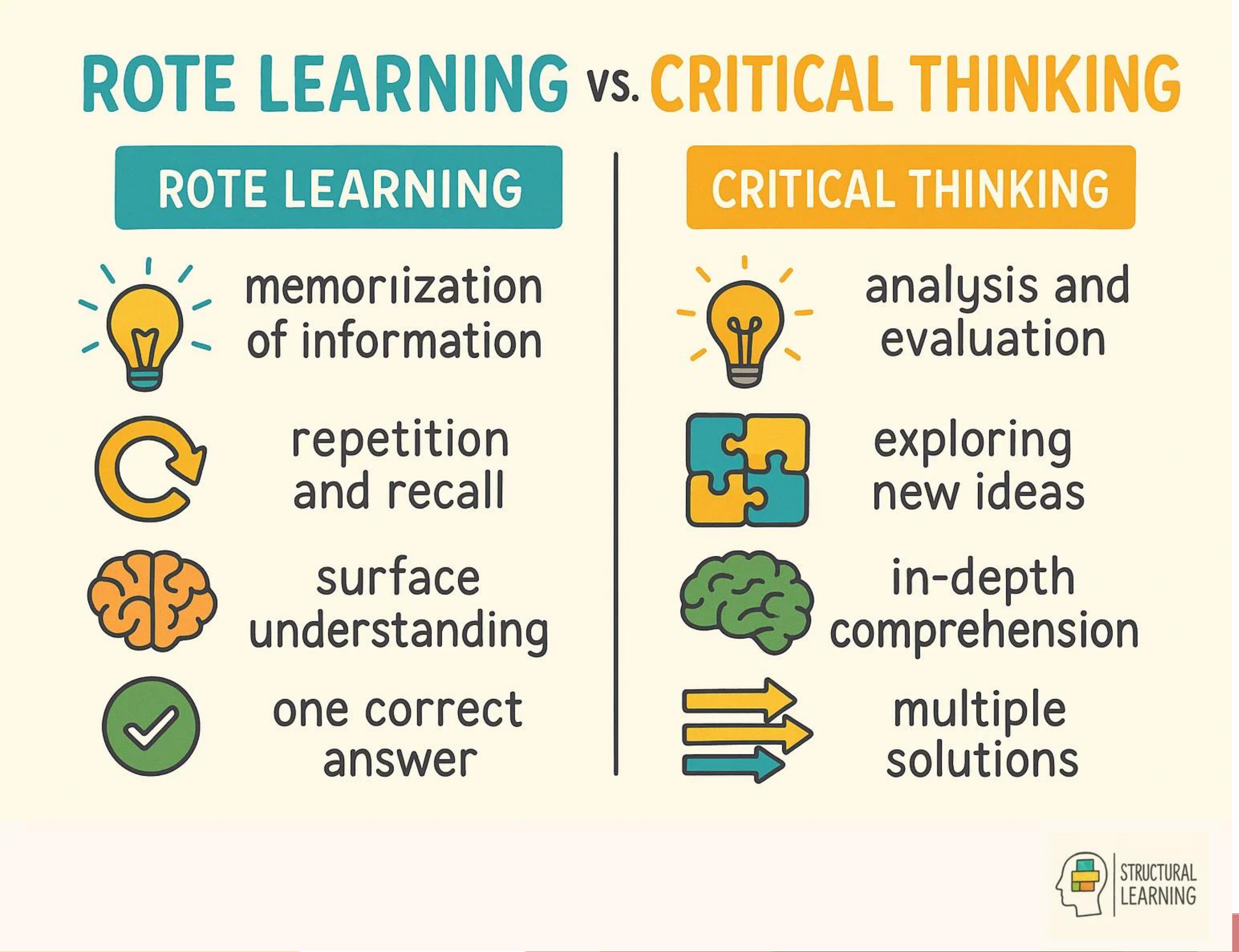
Assessment Areas:

- Analytical writing
- Group seminar presentation
- Trend impact analysis
- Exams (concept-based)

What This Course Is NOT?

Let’s set expectations clearly.

This course is NOT:
A programming class
A tool-training workshop
A memorization-heavy subject
A passive listening class



Final Thought: Your Role in the Future of IT

Technology does not shape the future alone.
People who design and govern it do.

Final Reflection:
You are future IT decision-makers
Your ethics will matter
Your analysis will influence systems
Your adaptability determines relevance